

信息检索与数据挖掘

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简要总结一下该文档

《Attention Is All You Need》文档总结

一、核心贡献

提出Transformer架构，这是首个完全基于注意力机制（摒弃循环和卷积结构）的序列 transduction 模型。它解决了传统循环神经网络（RNN）、长短期记忆网络（LSTM）等模型难以并行化、训练耗时久的问题，同时在序列建模任务中实现更优性能，为后续大语言模型发展奠定基础。

二、模型架构

1. 整体框架

遵循编码器 - 解码器结构，两者均由堆叠的自注意力层和逐点全连接前馈网络构成，且各子层均采用残差连接与层归一化，确保模型稳定训练，所有子层及嵌入层输出维度统一为 $d_{model} = 512$ 。

- 编码器：**由 6 个相同层堆叠而成，每层含“多头自注意力层 + 逐点全连接前馈网络”两个子层，可让输入序列各位置间自由建立依赖关系。
- 解码器：**同样由 6 个相同层堆叠，在编码器子层基础上新增“编码器 - 解码器注意力层”，且自注意

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Abstract

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The dominance of recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head



+ 新对话



→ Abstract

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《Attention Is All You Need》Abstract 解读

该摘要 (Abstract) 核心在于点明传统序列转换模型的局限、本文提出的创新架构及其实验成效与泛化能力, 具体内容可拆解为以下关键维度:

1. 传统模型背景

摘要首先指出, 当时主流的序列转换 (sequence transduction) 模型均基于复杂的循环神经网络 (recurrent neural networks) 或卷积神经网络 (convolutional neural networks), 这类模型普遍包含 “编码器 - 解码器” 结构; 即便性能最优的模型, 也需通过注意力机制 (attention mechanism) 连接编码器与解码器, 形成 “循环 / 卷积为主、注意力为辅” 的架构模式 ◆ 1-27。

2. 本文核心创新

针对传统模型的结构特点, 本文提出一种全新的简洁网络架构 —— **Transformer**。该架构完全依赖注意力机制构建, 彻底摒弃了传统模型中的循环结构与卷积结构, 是首个 “纯注意力驱动” 的序列转换模型, 从根本上改变了序列建模的核心依赖 ◆ 1-27。

3. 机器翻译实验验证

为验证 Transformer 的性能, 研究团队在两项 翻译任务上开展实验, 结果显示该模型具备三大显著优势:

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arXiv:1706.03762v7 [cs.CL] 2 Aug 2023

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Abstract

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The dominant sequence transduction models are recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

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2 Background (背景) 解读

文档的“2 Background”部分（对应◆1-38至◆1-42），核心是通过梳理“减少序列计算”的相关研究现状、自注意力机制的已有应用，明确 Transformer 与前人工作的关联与突破，为新架构的合理性提供上下文支撑，具体可拆解为以下四部分：

1. 减少序列计算的先行模型：卷积基模型的局限

背景部分首先指出，“减少序列式计算”是此前多个模型的核心设计目标，这类模型均以**卷积神经网络（CNN）**为基本构建块，通过并行计算所有输入 / 输出位置的隐藏表示，规避循环模型的“逐位置依赖”问题，典型代表包括：

- Extended Neural GPU（文献 [16]）
- ByteNet（文献 [18]）
- ConvS2S（文献 [9]）

但这些卷积基模型存在关键缺陷：**输入 / 输出序列中任意两个位置的信号关联，所需操作数随位置距**

离增加而增加。CNN 中操作数与距离呈“线性增长”，ByteNet（采用空洞卷积）中呈“对数增长”，Extended Neural GPU 中呈“平方增长”。

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states h_t , as a function of the previous hidden state h_{t-1} and the input for position t . This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [21] and conditional computation [32], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [16], ByteNet [18] and ConvS2S [9], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [12]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 27, 28, 22].

End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and language modeling tasks [34].

To the best of our knowledge, however, the Transformer is the first transduction model relying entirely on self-attention to compute representations of its input and output without using sequence-aligned RNNs or convolution. In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [17, 18] and [9].

3 Model Architecture

Most competitive neural sequence transduction models have an encoder-decoder structure [5, 2, 35]. Here, the encoder maps an input sequence of symbol representations $(x_1, \dots, x_n)$ to a sequence of continuous representations $\mathbf{z} = (z_1, \dots, z_n)$. Given $\mathbf{z}$, the decoder then generates an output sequence $(y_1, \dots, y_m)$ of symbols one element at a time. At each step the model is auto-regressive [10], consuming the previously generated symbols as additional input when generating the next.

创性，为后文详细介绍模型架构与实验结果建立了逻辑起点。

→ Figure 1: The Transformer - model architecture.

解释一下

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Figure 1: Transformer 模型架构解析

Figure 1 是《Attention Is All You Need》中核心的架构示意图，直观呈现了 Transformer 从输入到输出的完整流程，其核心是**对称的编码器 - 解码器栈结构**，图中左侧为编码器（Encoder）、右侧为解码器（Decoder），各组件的功能与连接逻辑需结合模型设计原理与文档描述（对应 ◆ 1-46、◆ 1-49、◆ 1-50、◆ 1-51）展开解读：

一、Figure 1 整体布局与核心定位

该图的核心作用是可视化 Transformer“完全基于注意力机制、无循环 / 卷积结构”的设计理念：

- 左侧**编码器栈**负责将输入序列（如源语言句子）映射为连续的上下文表示（ z 序列）；
- 右侧**解码器栈**负责基于编码器输出的上下文表示，逐 token 生成输出序列（如目标语言句子）；
- 两者均通过“堆叠相同层 + 残差连接 + 层归”实现深度建模，且所有组件的输出维度统一为 $d_{\text{model}} = 512$ ，确保模块间兼容 ◆ 1-46、◆ 1-50。

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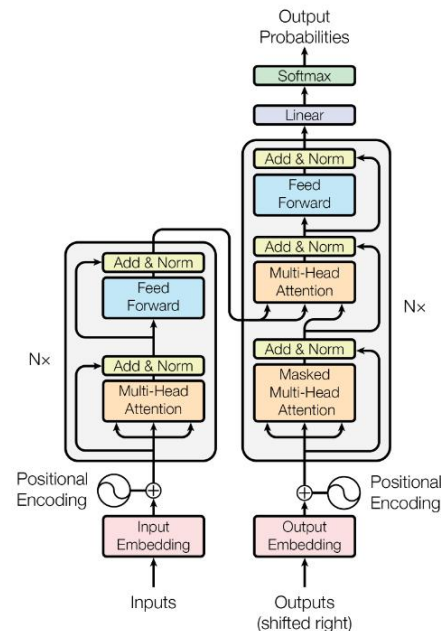


Figure 1: The Transformer - model architecture.

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The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder, shown in the left and right halves of Figure 1 respectively.

3.1 Encoder and Decoder Stacks

Encoder: The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-wise fully connected feed-forward network. We employ a residual connection [1] around each of the two sub-layers, followed by layer normalization [1]. That is, the output of each sub-layer is $\text{LayerNorm}(x + \text{Sublayer}(x))$, where $\text{Sublayer}(x)$ is the function implemented by the sub-layer itself. To facilitate these residual connections, all sub-layers in the model, as well as the embedding layers, produce outputs of dimension $d_{\text{model}} = 512$.

Decoder: The decoder is also composed of a stack of $N = 6$ identical layers. In addition to the two sub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head attention over the output of the encoder stack. Similar to the encoder, we employ residual connections around each of the sub-layers, followed by layer normalization. We also modify the self-attention sub-layer in the decoder stack to prevent positions from attending to subsequent positions. This

References

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《2017 Attention Is All You Need.pdf》References (参考文献) 解读

该论文的“References”部分（对应◆1-150至◆1-193）是全文研究的“学术支撑基石”，共收录40篇参考文献（编号[1]-[40]），覆盖**序列建模核心技术、注意力机制前身、优化与正则化方法、机器翻译基准模型**四大核心领域。这些文献不仅为Transformer的架构设计提供了理论依据与对比基准，也清晰展现了该研究与前人工作的继承和突破关系，具体可按功能拆解如下：

一、References的核心定位：学术严谨性与研究传承的体现

参考文献的核心作用是“锚定Transformer在学术脉络中的位置”：一方面，证明Transformer的创新并非“空中楼阁”——其关键组件（如层归一化、残差连接、注意力机制雏形）均建立在已验证的前沿研究基础上；另一方面，明确Transformer需对比的“传统架构标杆”（如RNN/LSTM、卷积序列模型），为实验结果的“优越性”提供参照。所有引用文献均为2017年及之前的领域关键成果，既包含经典奠基工作，也涵盖同期前沿探索，体现了作者对序列建模领域学术动态的全面把握◆1-150。

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Acknowledgements We are grateful to Nal Kalchbrenner and Stephan Gouws for their fruitful comments, corrections and inspiration.

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三、Figure 5 的核心意义：强化自注意力的结构建模能力与可解释性

Figure 5 并非孤立的可视化，而是对前文“自注意力优势”的补充与升华，其价值体现在两点：

1. 验证自注意力对句子结构的建模能力：传统循环 / 卷积层难以直接捕捉句子的层级结构（需通过多层堆叠间接学习），而 Figure 5 证明，自注意力头能直接关联句子的结构成分，无需依赖“逐步传递”或“层堆叠”——这种能力让 Transformer 在处理句法敏感任务（如 constituency 解析）时具备天然优势，呼应前文“Transformer 在英语 constituency 解析任务上泛化良好”的实验结果（◆ 1-141、◆ 1-202）。
2. 深化多头注意力的功能分工逻辑：Figure 3、4、5 共同构成“多头分工”的完整证据链——不同头分别负责“长距离短语依赖”“指代消解”“句子结构关联”等不同任务，这种自发的功能特化完美契合多头注意力的设计初衷（“联合关注不同子空间的依赖”，◆ 1-71）。同时，这种分工让模型的决策过程更易解释：研究者可通过注意力分布，明确看到“哪个头处理哪种结构”，而传统模型的内部机制难以如此直观拆解◆ 1-102、◆ 1-202。

综上，Figure 5 以“句子结构建模”为切入点，通过两个注意力头的实例，进一步证明了 Transformer 自注意力在句法语义建模上的优势，同时强化了“多头分工”的合理性与可解释性，为“全注意力架构优于传统循环 / 卷积架构”提供了更全面的可视化支撑。

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用自己的话解释一下Figure 5的具体内容 →

图5中的两个不同头具体是如何执行不同任务的？ →

Figure 5如何进一步补充说明了注意力头的功能分工？ →

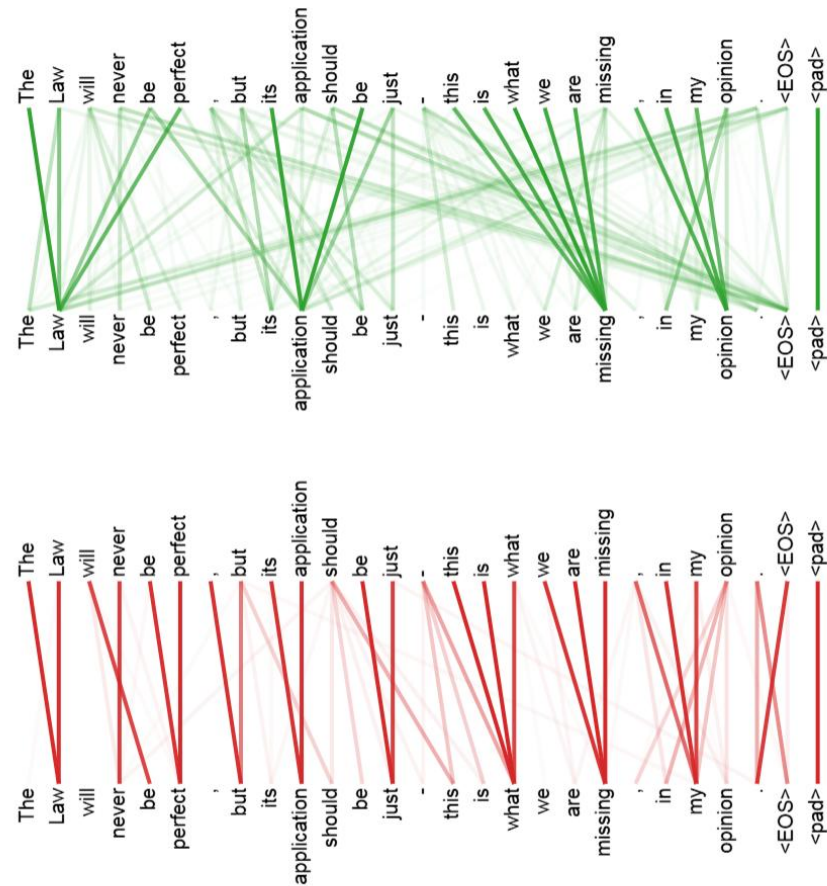


Figure 5: Many of the attention heads exhibit behaviour that seems related to the structure of the sentence. We give two such examples above, from two different heads from the encoder self-attention at layer 5 of 6. The heads clearly learned to perform different tasks.

《Attention Is All You Need》模型解读

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- **解码器**：同样由 6 个相同层堆叠，在编码器子层基础上新增“编码器 - 解码器注意力层”，且自注意力层通过掩码（屏蔽后续位置信息）保证自回归特性，[和豆包继续聊](#) 又依赖已生成位置信息。

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尹建华 1706 Attention Is All You Need

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豆包链接: https://www.doubao.com/thread/w27e29a2ebda9370d

论文PDF链接: https://arxiv.org/pdf/1706.03762

论文阅读笔记

1 问题陈述

飞书笔记链接：https://gwxyzj3n0h7.feishu.cn/wiki/BZWYwdHpPi1hLFkYbShcWk98n9b?from=from_copylink

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豆包链接: <https://www.doubao.com/thread/w27e29a2ebda9370d>

论文PDF链接: <https://arxiv.org/pdf/1706.03762>

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多模态推荐综述

扩散模型在推荐系统中的应用

学术搜索：Transformer与ChatGPT是什么关系？

基于 10 篇论文来源

Transformer 是一种神经网络架构，而 ChatGPT 是基于 Transformer 架构开发的一款大型语言模型，两者之间存在紧密且重要的关系，以下从多个方面深入阐述：

Transformer 架构的基础作用

核心架构支撑

多模态处理潜力

Transformer 对 ChatGPT 能力的塑造

强大的语言理解能力

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论文阅读笔记

1.问题陈述

作者要解决什么问题？如果论文的Introduction中有图例，截图放在笔记中。

2.方法概述

作者提出的方法是什么？如果论文有模型图或算法图，截图放在笔记中。

3 相关工作

使用LLM了解重要的相关工作，比如实验中选择的baselines，将总结性的内容，放在笔记中

论文阅读笔记

4.实验

作者使用了哪些数据集？数据集是否公开？链接是什么？

作者是否提供了代码？链接是什么？

实验的效果如何？截取论文的主要实验结果，并简单描述。

5.思考

论文的主要贡献是什么？

论文是否存在不足？具体有哪些？

对未来的研究方向有什么想法？

6.其它（选填）

需要特别记录的其它笔记